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MVJ COLLEGE OF ENGINEERING

Project title: AI-BASED GROUNDWATER POTENTIAL PREDICTION

Guide Name: Prof. Vishnu Mohan S

Abstract

Groundwater is essential for domestic, agricultural and industrial needs but its availability varies across regions due to environmental and terrain related factors. Conventional assessment methods can require extensive field surveys, making efficient regional prediction challenging. This project aims to develop an AI-based framework for predicting groundwater potential in the Bengaluru region of Karnataka.

The proposed approach uses machine learning to analyse groundwater conditions and their relationship with relevant environmental factors. LightGBM will be used to model nonlinear relationships and predict groundwater levels. Model performance will be evaluated using appropriate validation techniques and error metrics, while feature importance analysis will help identify influential factors.

The expected outcome is a groundwater potential map that can support groundwater exploration, regional planning and sustainable water resource management.

Base Paper 1: Machine Learning-Driven Groundwater Potential Prediction: A Data Driven Approach.

Base Paper 2: Prediction of the Groundwater Levels Based on Random Forest Regression Algorithm.

Team Members:

1. Abhilash V 1MJ24AI002
2. Kushal M 1MJ24AI027
3. Modeen Patel 1MJ24AI033
4. Prajwal Gowda P 1MJ24AI045